Stat 266 – Unit 8 – Video Set 5 – Class Examples

Summary:

What statistics are sufficient (good enough) to help formulate the MVUE?

Definition 9.3 : Sufficiency

The statistic $U = g(Y_1, \ldots, Y_n)$ is said to be sufficient for θ if the conditional distribution of $Y_1, \ldots, Y_n$, given U, does not depend on θ.

- Propose some U
- We also need the sampling distribution of U.

$\Rightarrow$ That is, if

$$L(\theta \mid U) = \begin{cases} \dfrac{L(\theta)}{p_U(u)}, & \text{for d.r.v.'s} \\ \dfrac{L(\theta)}{f_U(u)}, & \text{for c.r.v.'s} \end{cases}$$

doesn't depend on θ, then U is sufficient for θ.

Theorem 9.4 : Factorization

If $L(y_1, \ldots, y_n \mid \theta)$ can be factored into two non-negative functions,

$L(\theta) \Rightarrow \boxed{L(y_1, \ldots, y_n \mid \theta) = g(u, \theta) \times h(y_1, \ldots, y_n)}$

where $g(u, \theta)$ is a function only of u and θ and $h(y_1, \ldots, y_n)$ is not a function of θ, then $U = g(Y_1, \ldots, Y_n)$ is a sufficient statistic for θ.

Example A: Suppose $Y_1,\ldots,Y_n \overset{ind.}{\sim} Poisson(\lambda)$. Show that $U=\sum_{i=1}^{n} Y_i$ is sufficient for λ, by definition.

$$U=\sum_{i=1}^{n} Y_i \sim \text{Poisson}(n\lambda), \quad p_U(u)=\frac{(n\lambda)^u e^{-n\lambda}}{u!}, \quad u=0,1,2,\ldots$$

$$L(\lambda)=\prod_{i=1}^{n}\frac{\lambda^{y_i}e^{-\lambda}}{y_i!}=\frac{\lambda^{\sum y_i}e^{-n\lambda}}{\prod y_i!}=\frac{\lambda^{u}e^{-n\lambda}}{\prod y_i!}$$

$$L(\lambda\mid U)=\frac{L(\lambda)}{p_U(u)}=\frac{\cancel{\lambda^u}\cancel{e^{-n\lambda}}}{\prod y_i!}\times\frac{u!}{n^u\cancel{\lambda^u}\cancel{e^{-n\lambda}}}=\underbrace{\frac{u!}{n^u\prod y_i!}}_{\text{doesn't depend on }\lambda.}$$

$U=\sum_{i=1}^{n} Y_i$ is sufficient for λ.

Also, $\bar{Y}$ is sufficient for λ and unbiased.

<u>Note:</u> We only need <u>one</u> statistic, so we say $\bar{Y}$ (or $\sum Y_i$) is minimally sufficient for λ.

Unbiased estimators as functions of minimally sufficient statistics will be our MVUE.

So $\bar{Y}$ is the MVUE for $\mu_Y=\lambda$.

$$L(\lambda) = \frac{\lambda^{\sum y_i} e^{-n\lambda}}{\prod_{i=1}^{n} y_i!} = g(u, \lambda) \times h(y_1, \ldots, y_n)$$

$$= \underbrace{\lambda^{\sum y_i} e^{-n\lambda}}_{g(u,\lambda)} \times \underbrace{\frac{1}{\prod y_i!}}_{h(y_1,\ldots,y_n)} = \underbrace{\lambda^{n\bar{y}} e^{-n\lambda}}_{\bar{y} \text{ is minimally sufficient.}} \times \frac{1}{\prod y_i!}$$

So $U = \sum_{i=1}^{n} Y_i$ is minimally sufficient for λ.

only one statistic

Note: $$L(\lambda) = \underbrace{\frac{\lambda^{\sum y_i} e^{-n\lambda}}{\prod y_i!}}_{g(u_1, u_2, \lambda)} \times \underbrace{1}_{h(y_1,\ldots,y_n)}$$

$U_1 = \sum_{i=1}^{n} Y_i$ and $U_2 = \prod_{i=1}^{n} Y_i!$ are jointly sufficient for λ. But not minimally sufficient. We don't actually need U_2.

Example B: Suppose $Y_1,\ldots,Y_n \overset{ind.}{\sim} Gamma(\alpha,\beta)$. Find sufficient statistics for α and β.

$$f_Y(y) = \frac{y^{\alpha-1} e^{-y/\beta}}{\Gamma(\alpha)\beta^{\alpha}}, \quad y > 0$$

From a sample $Y_1 = y_1, Y_2 = y_2, \ldots, Y_n = y_n$,

$$L(\alpha,\beta) = \prod_{i=1}^{n} f_Y(y_i) = \prod_{i=1}^{n} \frac{y_i^{\alpha-1} e^{-y_i/\beta}}{\Gamma(\alpha)\beta^{\alpha}}$$

$$= \underbrace{\frac{\left(\prod_{i=1}^{n} y_i\right)^{\alpha-1} e^{-\frac{1}{\beta}\sum^{n} y_i}}{(\Gamma(\alpha))^n \beta^{\alpha n}}}_{g(u_1,u_2,\alpha,\beta)} \times \underbrace{1}_{h(y_1,\ldots,y_n)}$$

So $U_1 = \prod_{i=1}^{n} Y_i$ and $U_2 = \sum_{i=1}^{n} Y_i$ are jointly and minimally sufficient for α and β.

1) Suppose α is a known constant:

$$L(\beta) = \underbrace{\frac{1}{\beta^{\alpha n}} e^{-\frac{1}{\beta}\sum y_i}}_{g(u,\beta),} \times \underbrace{\frac{(\prod y_i)^{\alpha-1}}{[\Gamma(\alpha)]^n}}_{h(y_1,\dots,y_n)}$$

$U = \sum_{i=1}^{n} Y_i$ is minimally sufficient for β.

2) Suppose β is a known constant:

$$L(\alpha) = \underbrace{\frac{(\prod y_i)^{\alpha-1}}{\beta^{\alpha n}[\Gamma(\alpha)]^n}}_{g(u,\alpha)} \times \underbrace{e^{-\frac{1}{\beta}\sum y_i}}_{h(y_1,\dots,y_n)}$$

$U = \prod_{i=1}^{n} Y_i$ is minimally sufficient for β.

$Y \sim \text{Gamma}(\alpha, \beta)$

<u>MME:</u> • $\mu_Y = \alpha\beta \Rightarrow m_1' = \bar{Y}$ $\qquad (\alpha\beta = \bar{Y})$

• $E[Y^2] = \sigma_Y^2 + \mu_Y^2 = \alpha\beta^2 + (\alpha\beta)^2 \Rightarrow m_2' = \frac{1}{n}\sum Y_i^2$

$$\left(\alpha\beta^2 + \alpha^2\beta^2 = \frac{1}{n}\sum Y_i^2\right)$$

• Solve for α and β to find $\hat{\alpha}_{MM}$ and $\hat{\beta}_{MM}$.

<u>MLE:</u> $L(\alpha, \beta) = \dfrac{(\prod y_i)^{\alpha-1} e^{-\frac{1}{\beta}\sum y_i}}{(\Gamma(\alpha))^n \beta^{\alpha n}}$

• $\ell(\alpha, \beta) = \ln(L(\alpha, \beta)) = (\alpha - 1)\ln(\prod y_i) - \frac{1}{\beta}\sum y_i$

$$- n\ln(\Gamma(\alpha)) - \alpha n \ln\beta.$$

• $\dfrac{\partial \ell(\alpha, \beta)}{\partial \beta} = \dfrac{\sum y_i}{\beta^2} - \dfrac{\alpha n}{\beta} \Rightarrow 0$, $\beta = \dfrac{\sum Y_i}{\alpha n} = \dfrac{\bar{Y}}{\alpha}$

• $\dfrac{\partial \ell(\alpha, \beta)}{\partial \alpha} = \ln(\prod y_i) - n\dfrac{d}{d\alpha}\underbrace{(\ln(\Gamma(\alpha)))}_{?} - n\ln\beta$

Example C: $Y_1,\ldots,Y_n \overset{ind.}{\sim} f_Y(y) = \begin{cases} e^{-(y-\theta)} & , y > \theta \\ 0 & , \text{else} \end{cases}$

$$L(\theta) = \prod_{i=1}^{n} f_Y(y_i) = \prod_{i=1}^{n} e^{-y_i} e^{\theta} = e^{-\sum y_i} e^{n\theta}$$

$$\neq \underbrace{e^{n\theta}}_{g(u,\theta)} \times \underbrace{e^{-\sum y_i}}_{h(y_1,\ldots,y_n)} \quad , \text{ all } y_i > \theta$$

no statistic here $\Rightarrow$ no U.

Indicator variables: Define

$$I_{(a,b)}(y) = \begin{cases} 1, & \text{if } a < y < b \\ 0, & \text{otherwise} \end{cases}.$$

$$f_Y(y) = \begin{cases} e^{-(y-\theta)} & , y > \theta \\ 0 & , \text{else} \end{cases} = e^{-(y-\theta)} I_{(\theta,\infty)}(y)$$

$$f_Y(y) = e^{-(y-\theta)} I_{(\theta,\infty)}(y)$$

$$L(\theta) = \prod_{i=1}^{n} e^{-y_i} e^{\theta} I_{(\theta,\infty)}(y_i)$$

$$= e^{-\sum y_i} e^{n\theta} \underbrace{\prod_{i=1}^{n} I_{(\theta,\infty)}(y_i)}$$

$= 1$ iff all $y_i > \theta$

$= 1$ iff $y_{(1)} > \theta$

$$= \underbrace{e^{n\theta} I_{(\theta,\infty)}(y_{(1)})}_{g(u,\theta)} \times \underbrace{e^{-\sum y_i}}_{h(y_1,\dots,y_n)}$$

$\Rightarrow U = Y_{(1)}$ is minimally sufficient for θ.

<u>Add-in:</u> Consider $Y_1, \ldots, Y_n \overset{iid}{\sim} \text{Uniform}(\underbrace{a, b}_{\text{Parameters.}})$.

$$f_Y(y) = \begin{cases} \frac{1}{b-a}, & a < y < b \\ 0, & \text{else} \end{cases} = \frac{1}{b-a} \underbrace{I_{(a,b)}(y)}_{= \begin{cases} 1, & a < y < b \\ 0, & \text{else} \end{cases}}$$

$$L(a, b) = \prod_{i=1}^{n} f_Y(y_i) = \prod_{i=1}^{n} \left(\frac{1}{b-a}\right) I_{(a,b)}(y_i)$$

$$= \left(\frac{1}{b-a}\right)^n \prod_{i=1}^{n} I_{(a,b)}(y_i)$$

$$= \underbrace{\left(\frac{1}{b-a}\right)^n I_{(a,\infty)}(y_{(1)}) \, I_{(-\infty,b)}(y_{(n)})}_{g(u_1, u_2, a, b)} \times \underbrace{1}_{h(y_1, \ldots, y_n)}$$

So $Y_{(1)}$ and $Y_{(n)}$ are jointly and minimally sufficient for a and b.

Suppose

$$L(a,b) = \left(\frac{1}{b-a}\right)^n I_{(a,b)}(y_{(1)})\, I_{(a,b)}(y_{(n)}).$$

What if we know $a = 0$: (known constant)

$$L(b) = \left(\frac{1}{b}\right)^n I_{(0,b)}(y_{(1)})\, I_{(0,b)}(y_{(n)})$$

not needed

$$= \left(\frac{1}{b}\right)^n I_{(0,\infty)}(y_{(1)})\, I_{(-\infty,b)}(y_{(n)})$$

$$= \underbrace{\left(\frac{1}{b}\right)^n I_{(-\infty,b)}(y_{(n)})}_{g(u,b)} \times \underbrace{I_{(0,\infty)}(y_{(1)})}_{h(y_1,\ldots,y_n)}.$$

So $U = Y_{(n)}$ is minimally sufficient for b.

Example D:

Suppose $Y_1,\ldots,Y_n$ is a random sample of independent and identically distributed random variables with density function given by

$$f_Y(y)=\begin{cases}(1-\beta)y^{-\beta} & ,\ 0<y<1,\ \ \beta<1\\ 0 & ,\ \text{else}\end{cases}.$$

What is sufficient for β?

$$\hat{\beta}_{MM} = \frac{1-2\bar{Y}}{1-\bar{Y}}$$

$$\hat{\beta}_{MLE} = 1 + \frac{n}{\ln(\prod Y_i)}.$$

$$L(\beta) = \prod_{i=1}^{n}(1-\beta)y_i^{-\beta} = \underbrace{(1-\beta)\left(\prod y_i\right)^{-\beta}}_{g(u,\beta)} \times \underbrace{1}_{h(y_1,\ldots,y_n)}$$

So $U = \prod_{i=1}^{n} Y_i$ is minimally sufficient for β.

Example E:

Suppose $Y_1,\ldots,Y_n$ is a random sample of independent and identically distributed random variables with density function given by

$$f_Y(y)=\begin{cases}\dfrac{\alpha\beta^{\alpha}}{y^{\alpha+1}}, & 0<\underline{\beta<y},\alpha>0\\ 0 & \text{, else}\end{cases}. \quad = \frac{\alpha\beta^{\alpha}}{y^{\alpha+1}} I_{(\beta,\infty)}(y)$$

What statistics are jointly sufficient for α and β?

$$L(\alpha,\beta) = \prod_{i=1}^{n} f_Y(y_i) = \prod_{i=1}^{n} \frac{\alpha\beta^{\alpha}}{y_i^{\alpha+1}} I_{(\beta,\infty)}(y_i)$$

$$= \frac{\alpha^n\beta^{\alpha n}}{\left(\prod^{n} y_i\right)^{\alpha+1}} \prod_{i=1}^{n} I_{(\beta,\infty)}(y_i)$$

$$= \underbrace{\frac{\alpha^n\beta^{\alpha n}}{\left(\prod^{n} y_i\right)^{\alpha+1}} I_{(\beta,\infty)}(y_{(1)})}_{g(u_1,u_2,\alpha,\beta)} \times \underbrace{1}_{h(y_1,\ldots,y_n)}$$

$U_1 = \prod_{i=1}^{n} Y_i$ and $U_2 = Y_{(1)}$ are jointly and minimally sufficient for α and β.

Recall:

MM: $\hat{\beta}_{MM}$ and $\hat{\alpha}_{MM}$ are functions of $\mu_1' = \bar{Y}$ and $\mu_2' = \frac{1}{n}\sum Y_i^2$.

MLE: $\hat{\beta}_{MLE} = Y_{(1)}$

$$\underbrace{\hat{\alpha}_{MLE} = \frac{n}{\ln\left(\prod^{n} Y_i\right) - n\ln Y_{(1)}}}_{\text{Functions of the minimally sufficient statistics.}}$$